Completeness, Supremum, Dedekind Cuts, and Metric Spaces

N-20220713

§1.1 Order, bounds, and the least-upper-bound property

A simple but important fact is that for every positive real A, there is a positive real $M > A$, and for every positive real a, there is a positive real $\varepsilon < a$. For instance,

$$M = A + 1, \qquad \varepsilon = \frac{a}{2}.$$

This is the first appearance of the ε-language of analysis. Our treatment follows the ordered-field and metric-completion development in the MIT 18.100A notes and Jiří Lebl's *Basic Analysis I*; see especially the discussion of supremum and Cauchy completeness.[1]

We think of $\mathbb{R}$ as a line, where order is represented by left and right. This picture is helpful, but the note emphasizes that all conclusions must ultimately come from axioms and rigorous reasoning, not from the drawing itself.

A set $X \subseteq \mathbb{R}$ is bounded above if there exists $A \in \mathbb{R}$ such that

$$x \leq A \quad \text{for all } x \in X.$$

Such an A is an upper bound. Similarly, X is bounded below if there exists a such that

$$a \leq x \quad \text{for all } x \in X.$$

The least upper bound property says: if $X \subseteq \mathbb{R}$ is nonempty and bounded above, then the set of upper bounds has a least element. This least upper bound is denoted

$$\sup X.$$

Equivalently, $M = \sup X$ iff:

(i) $x \leq M$ for all $x \in X$;

(ii) for every $\varepsilon > 0$, there exists $x \in X$ such that

$$x > M - \varepsilon.$$

The note sketches a proof of the supremum property from the nested interval principle by bisecting intervals and selecting the half that still contains upper bounds or points of the set.

The dual notion is the infimum:

$$\inf X.$$

One always has

$$\inf X \leq \sup X$$

[1] MIT 18.100A, *Real Analysis*, lecture notes.

for a nonempty bounded set.

The supremum of a set $A \subseteq \mathbb{R}$ is not merely a maximum. It is the least number s such that every element of A is at most s. A maximum must belong to A; a supremum need not.

For example,

$$\sup(0,1) = 1,$$

but $(0,1)$ has no maximum. Completeness has a concrete obstruction over $\mathbb{Q}$. The set $S = \{q \in \mathbb{Q} : q > 0,\ q^2 < 2\}$ is nonempty and bounded above but has no supremum in $\mathbb{Q}$; any putative rational boundary would satisfy $r^2 = 2$, contradicting the elementary parity argument that $\sqrt{2}$ is irrational. Therefore the ordered-field axioms alone do not imply the supremum property.

§1.2 Completeness and nested-interval bisection

Let $X \subseteq \mathbb{R}$ be nonempty and bounded above. One can construct nested intervals

$$I_n = [a_n, b_n]$$

so that a_n is not an upper bound of X, while b_n is an upper bound, and

$$b_n - a_n \leq 2^{-n}.$$

The nested interval axiom gives a point

$$M \in \bigcap_{n=1}^{\infty} I_n.$$

At each bisection, retain the half whose lower endpoint is not an upper bound and whose upper endpoint is an upper bound; the interval length halves. The nested-interval property then supplies M. The shrinking length forces M to be the least upper bound of X. If M were not an upper bound, some $x \in X$ would lie above it; for large n, the right endpoint b_n would be too close to M, contradiction. If M were not least, a smaller upper bound would contradict the construction of the a_n's.

The supremum property says every nonempty bounded-above set has a least upper bound. This is the order-theoretic heart of the real line. It is what makes monotone convergence and intermediate-value reasoning possible.

§1.3 Dedekind cuts and the order completion

A Dedekind cut is a subset $X \subset \mathbb{Q}$ such that:

(i) $X \neq \varnothing$ and $\mathbb{Q} \setminus X \neq \varnothing$;

(ii) if $x \in X$ and $y < x$, then $y \in X$;

(iii) X has no greatest element.

A rational number p/q defines the cut

$$X_{p/q} = \{x \in \mathbb{Q} : x < p/q\}.$$

An irrational such as $\sqrt{2}$ is represented by

$$X_{\sqrt{2}} = \{x \in \mathbb{Q}_{>0} : x^2 < 2\} \cup \mathbb{Q}_{\leq 0}.$$

This is an actual lower cut: it is nonempty, proper, downward closed and has no greatest rational element. The last property uses density of the rationals. The cut does not come from any rational endpoint, because 2 is not a square in $\mathbb{Q}$. Thus real numbers can be constructed from rational cuts.

A Dedekind cut separates rationals into a lower part and an upper part with no gap. Irrational numbers appear as cuts not produced by a rational boundary. Completeness says that every such cut corresponds to a real number.

This is the philosophical point of constructing $\mathbb{R}$: the real line is obtained by filling all rational gaps.

Dedekind cuts construct reals by filling order gaps; Cauchy completion constructs reals by adding missing limits of Cauchy sequences. The two constructions produce isomorphic complete ordered fields, but they emphasize different structures: order versus metric.

§1.4 Metric spaces, dense subspaces, and Cauchy completion

A metric space is a set X equipped with a function

$$d : X \times X \to \mathbb{R}_{\geq 0}$$

satisfying:

(i) $d(x, y) = 0$ iff $x = y$;

(ii) $d(x, y) = d(y, x)$;

(iii) $d(x, z) \leq d(x, y) + d(y, z)$.

The usual metric on $\mathbb{R}$ is

$$d(x, y) = |x - y|.$$

On $\mathbb{R}^q$, common metrics are:

$$d_2(x, y) = \left(\sum_{n=1}^{q} |x_n - y_n|^2 \right)^{1/2},$$

$$d_1(x, y) = \sum_{n=1}^{q} |x_n - y_n|,$$

and

$$d_\infty(x, y) = \max_{1 \leq n \leq q} |x_n - y_n|.$$

The note remarks that d_2 requires Cauchy–Schwarz for the triangle inequality, d_1 is the Manhattan metric, and d_∞ is often called the Chebyshev metric.

A final example is the discrete metric:

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

In the discrete metric, open balls are especially simple:

$$B_r(x) = \begin{cases} \{x\}, & 0 < r \leq 1, \\ X, & r > 1. \end{cases}$$

Boundary points are not included in an open ball, which explains why the case $r = 1$ gives $\{x\}$, not all of X.

If $Y \subseteq X$ and X has metric d, then Y inherits the subspace metric

$$d_Y(y_1, y_2) = d(y_1, y_2).$$

A subset $Y \subseteq X$ is dense if for every $x \in X$ and every $\varepsilon > 0$, there exists $y \in Y$ such that

$$d(x, y) < \varepsilon.$$

This formalizes the idea that Y comes arbitrarily close to every point of X.

A metric space can be completed by adding equivalence classes of Cauchy sequences. Two Cauchy sequences are equivalent if their distance tends to zero. The real numbers can be constructed as the completion of $\mathbb{Q}$ under the usual metric. For example, set $q_n = \lfloor 10^n \sqrt{2} \rfloor / 10^n$. Each q_n is rational and $|q_n - \sqrt{2}| < 10^{-n}$; hence (q_n) is Cauchy in $\mathbb{Q}$, but has no rational limit. The completion adjoins its equivalence class. Two Cauchy sequences represent the same real number precisely when their termwise difference tends to zero. This turns the philosophical claim about "filling gaps" into an explicit construction.